

The gradient of $s - t$ graph gives the velocity v . answer is C.

- 5 D** Let F be the upward force by $3m$ on the system of $(m + 5m)$.

$$F_{\text{net}} \text{ on system of } (m + 5m) = F - 6mg$$

$$F - 6mg = 6ma$$

$$F = 6mg + 6ma$$

$$\text{By Newton's 3}^{\text{rd}} \text{ law, magnitude of force } m \text{ exerts on } 3m = F = 6mg + 6ma$$

- 6 A** Volume of sphere $= M/\rho = 0.75/650 = 1.15 \times 10^{-3} \text{ m}^3$